

ROBOSOC CER

High-Current Direct RC BTS7960 Drivetrain & SG90 Servo Potentiometer Steering Integration

COLLEGE ENGINEERING PROJECT

HARDWARE & CIRCUIT DESIGN

DIRECT RC ROBOTICS

1. OVERVIEW

Project Objective: The primary objective of this project is to build and configure a high-torque, microcontroller-less 4-Wheel Drive (4WD) Robo-Soccer robot. The system is designed to achieve low latency, maximum efficiency, and high reliability by controlling the high-power BTS7960 motor drivers directly using the 2.4GHz FlySky receiver and custom feedback elements from extracted SG90 servo potentiometers.

Project Importance: Eliminating microcontrollers (like Arduino) removes code failure points, boot delays, and processing latency. This pure hardware-interfaced RC approach demonstrates fundamental analog and PWM control engineering:

- Structural Durability:** The custom **Iron Chassis** provides maximum weight, stability, and impact tolerance required for competitive arena battles.
- High-Traction Wheels:** Custom **3D-printed PLA wheels with rubber strip treads** deliver superior grip on soccer mats.
- Direct RC Control:** The **FlySky Receiver** directly signals the dual 43A **BTS7960 Motor Drivers** without intermediary code processing.
- Custom Feedback & Steering Modification:** Re-purposing the internal **SG90 servo potentiometer** provides exact manual mechanical feedback and position tuning for mechanical mechanisms.
- Power Regulation:** A **Digital Display Buck Converter** steps down the high-discharge **3S 2200mAh 80C LiPo battery** voltage to safely power the receiver while offering real-time voltage monitoring.

2. SYSTEM COMPONENTS & HARDWARE BILL OF MATERIALS

Subsystem	Component Specification	Qty	Engineering Justification
Chassis Structure	Heavy-Duty Custom Iron Frame	1	Provides extreme structural stiffness and high mass to withstand intense robot collisions during gameplay.
Drive Wheels	PLA 3D-Printed Wheels with Rubber Strips	4	Lightweight rim structures with flexible rubber treads for maximum coefficient of friction on soccer mats.
Actuators	12V Johnson DC Geared Motors	4	Delivers high continuous torque and acceleration required for rapid dynamic manoeuvres and pushes.
Motor Controllers	BTS7960 43A High-Power H-Bridge Drivers	2	Handles massive surge currents up to 43A, directly driven by RC receiver PWM channels without microcontrollers.
Primary Power	3S LiPo Battery (11.1V, 2200mAh, 80C)	1	Delivers high continuous current (2200mA $\times$ 80 = 176A peak burst) to satisfy high torque demands.
Voltage Regulator	Buck Converter with Digital Voltmeter Display	1	Steps down 11.1V to a stable 5V for the FlySky RX while offering real-time LED battery monitoring.
RC System			

Subsystem	Component Specification	Qty	Engineering Justification
	FlySky FS-i6 TX & FS-iA6B / FS-R6B RX	1 set	Provides direct 2.4GHz proportional PWM signal channels to drive high-power motor drivers directly.
Position Sensing	Extracted SG90 Servo Potentiometers	1-2	Salvaged mini-potentiometers used for fine-tuning signal thresholds and mechanical feedback position.

3. DIRECT CIRCUIT WIRING & HARDWARE INTERFACE

CRITICAL POWER SAFETY NOTE: COMMON GROUND & VOLTAGE ISOLATION

Since no microcontroller is used, the **FlySky Receiver** gets its 5V power exclusively from the **Digital Display Buck Converter**. Ensure the **LiPo Power Negative (-)**, **Buck Converter GND**, **BTS7960 Logic GND**, and **Receiver GND** are connected together to form a common ground reference.

3.1 High-Current Power Wiring

- **Battery Power Connection:** Connect the 3S 2200mAh 80C LiPo (+11.1V / +12.6V fully charged) to the high-current distribution board.
- **BTS7960 Power Inputs:** Wire LiPo (+) to **B+** and LiPo (-) to **B-** on both BTS7960 drivers. Connect the left-side Johnson Motors in parallel to **M+/M-** on Driver 1, and right-side Johnson Motors in parallel to **M+/M-** on Driver 2.
- **Buck Converter Setup:** Wire the input of the Buck Converter to the LiPo battery terminals. Turn the adjustment screw until the integrated display reads exactly **5.00V DC**, then connect the 5V output to power the FlySky receiver VCC pin.

3.2 Receiver Direct Pin Mapping to BTS7960 Motor Drivers

In a direct RC drive setup, PWM channels from the FlySky receiver are connected directly to the control inputs of the BTS7960 driver modules:

FlySky Receiver Channel	Target Component Signal Pin	Function / Movement
Channel 1 (PWM Signal)	Driver 1 (RPWM / LPWM Inputs)	Controls Left Side Johnson Motors (Forward / Reverse)
Channel 2 (PWM Signal)	Driver 2 (RPWM / LPWM Inputs)	Controls Right Side Johnson Motors (Forward / Reverse)
VCC / GND (+5V Rail)	BTS7960 VCC / GND & Buck Converter 5V Out	Supplies power to receiver logic and driver enable pins (R_EN / L_EN tied HIGH)

3.3 Extracted SG90 Potentiometer Wiring & Mechanical Tuning

- **Potentiometer Terminals:** The three wires desoldered from the SG90 micro servo potentiometer correspond to **VCC (5V)**, **GND**, and **Wiper Signal**.
- **Direct Integration:** The wiper signal is tied into the custom feedback/steering link or used as a variable resistive divider to adjust zero-point neutral trim physically on the robot chassis.

4. HARDWARE TUNING & FIELD CALIBRATION

- **Transmitter Mix Setup:** Set up **Elevon / V-Tail mixing** directly on the FlySky FS-i6 transmitter settings menu. This allows single-stick (arcade) steering control without needing software mixing code.
- **Zero-Point Trim Adjustment:** Use the extracted SG90 potentiometer fine-tuning dial to ensure motors remain completely stationary when joystick controls are at neutral.
- **Voltage Monitoring:** Keep an eye on the digital display of the buck converter during matches; if input voltage drops below 9.9V (3.3V per cell), swap out the LiPo battery to avoid deep discharge damage.
- **Traction & Chassis Maintenance:** Inspect the rubber strip treads on the 3D-printed PLA wheels and clean them with isopropyl alcohol to maintain maximum arena grip.

5. CONCLUSION

Project Summary & Engineering Conclusion

This 4WD Robo-Soccer robot demonstrates a high-performance, direct RC hardware architecture that eliminates software lag and microcontroller overhead. By pairing high-torque **12V Johnson motors** with **BTS7960 43A motor drivers** and an **80C high-discharge 3S LiPo battery**, the drivetrain delivers maximum torque and instant throttle response.

The combination of a heavy-duty **Iron Chassis**, high-friction **PLA 3D-printed wheels with rubber treads**, onboard **digital display voltage regulation**, and modified **SG90 potentiometer feedback tuning** creates a reliable, robust, and competition-ready soccer bot built specifically for hardware-focused engineering tournaments.